

The Science of Seeing

● Green Level: I Want More Support

Name: _____ Date: _____ Period: _____

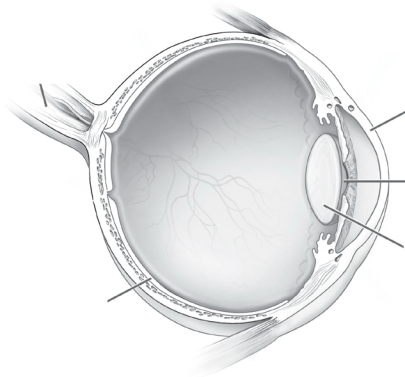
How Your Eyes Work

Your eyes collect light, but your **brain** creates the picture you see! Light travels through your eye and gets turned into electrical signals that your brain interprets.

Label the Eye

Use the **Word Bank** to label the parts of the eye diagram below.

Word Bank: Cornea • Pupil • Lens • Retina • Optic Nerve



1. _____ (clear front part, bends light)
2. _____ (black circle, controls light amount)
3. _____ (focuses light onto back of eye)
4. _____ (light-sensitive layer with rods and cones)
5. _____ (carries signals to brain — location of your blind spot!)

The Blind Spot Experiment

Use the Blind Spot Test Card your teacher gave you. Follow these steps:

1. Close your **LEFT** eye
2. Look at the + with your **RIGHT** eye
3. Slowly move the card closer to your face
4. The dot should disappear!

Why does this happen? Circle the correct answer:

A) The dot moves off the card B) Your optic nerve has no light receptors (blind spot) C) You blinked

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Two Types of Illusions

Eye-based illusions: Trick the cells in your eye (like afterimages that tire out your cones)

Brain-based illusions: Trick how your brain interprets what it sees (like size illusions)

Match the Illusion!

Draw a line from each illusion to where it happens:

Color Afterimage

EYE (retina)

Size Illusion

BRAIN (visual cortex)

Blind Spot

Mini Museum Connection

The **stereoscope** in the Mini Museum tricks your brain! It shows each eye a slightly different picture. Your brain combines them into one 3D image.

Try the Perception Lab: minimuseumproject.com/exhibitions/seeing-is-deceiving/illusions

What I Learned

Fill in the blanks:

My eyes collect _____, but my _____ creates the picture I see.

The blind spot is where the _____ connects to my eye.

Optical illusions work by tricking either my _____ or my _____.

Share with Your Family!

Try the blind spot experiment with your family! Can they find their blind spot too?

Fun fact to share: Your blind spot is about the size of 17 full moons in the sky—but you never notice it because your brain fills it in!